

Lightweight AI-Driven Multi-Objective Optimization for Renewable and Carbon Aware Cloud Scheduling

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Abstract—The exponential expansion of cloud computing infrastructures has significantly increased global energy consumption, contributing to elevated carbon emissions and environmental degradation. To address these challenges, this paper presents a framework for renewable and carbon aware cloud scheduling. The proposed system integrates historical carbon intensity and meteorological datasets—including parameters such as cloud cover, wind speed, and temperature—collected from multiple geographic regions. Using Random Forest-based time series forecasting, the framework predicts future energy conditions and renewable availability with high accuracy.

Subsequently, a multi-objective optimization strategy based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is employed to determine optimal job schedules that minimize carbon intensity, maximize renewable energy utilization, and ensure effective load balancing across regions. The framework operates with a lightweight computational footprint, making it suitable for real-time deployment in distributed cloud environments.

Experimental results demonstrate that the proposed approach effectively reduces average carbon intensity while maintaining performance and equitable workload distribution. This work contributes to the development of sustainable, carbon-aware cloud scheduling mechanisms, aligning with global initiatives such as UN Sustainable Development Goals (SDG 7 – Affordable and Clean Energy, and SDG 13 – Climate Action).

Index Terms—Cloud scheduling, multi-objective optimization, renewable energy, carbon intensity, NSGA-II, machine learning, green computing, sustainable cloud infrastructure, AI-based scheduling, carbon-aware systems.

I. INTRODUCTION

The exponential growth of cloud computing has transformed the global digital ecosystem by enabling scalable, on-demand computing resources for diverse applications. However, this advancement has also resulted in a sharp increase in energy consumption, contributing significantly to global carbon emissions. Modern data centers now represent one of the largest consumers of electricity, much of which is still generated from fossil fuels. Consequently, there is an urgent need to design sustainable cloud scheduling mechanisms that optimize not only performance and cost but also environmental impact.

Conventional cloud schedulers primarily focus on optimizing computational efficiency, latency, and throughput, often disregarding the carbon intensity of the power sources that sustain their operations. As the world transitions toward renewable energy, variability in renewable generation—driven by changing weather and geographical conditions—poses additional challenges to effective scheduling. To address this,

recent research has explored carbon-aware and renewable-integrated scheduling approaches that intelligently align cloud workloads with periods of low carbon intensity and high renewable energy availability [7], [22], [23].

Yet, existing solutions are often computationally expensive, lack real-time adaptability, or fail to balance sustainability objectives with operational performance. In this context, the proposed work introduces a Lightweight AI-Driven Multi-Objective Optimization Framework for Renewable and Carbon-Aware Cloud Scheduling that integrates short-term forecasting and multi-objective evolutionary optimization.

II. LITERATURE SURVEY

Cloud and data-center energy research has increasingly shifted from mere efficiency improvements to explicit carbon-awareness, driven by the recognition that scheduling decisions can materially affect greenhouse-gas emissions when electricity carbon intensity and renewable availability vary over time and geography. Several recent studies develop forecasting systems that predict grid carbon intensity at day-ahead and multi-day horizons, demonstrating that short-term forecasts can be accurate enough to guide workload timing and placement decisions [1], [3], [13].

Parallel to forecasting advances, a body of work has focused on renewable-aware and carbon-aware scheduling algorithms that shift workloads in time (temporal shifting) and space (geographic shifting) to exploit low-carbon periods and regions. Empirical and simulation studies indicate that spatial-temporal shifting can yield significant emission reductions for delay-tolerant and batch workloads, though effectiveness depends on workload elasticity, prediction accuracy, and the cost/latency trade-offs of migration [22], [23]. Reviews of carbon-aware scheduling techniques underline that many approaches do either spatial or temporal shifting, while fewer integrate both forecasting and multi-region decision-making in a unified pipeline—an integration that enables finer-grained exploitation of renewable availability [12].

Machine learning plays a central role across forecasting and scheduling research. Traditional statistical methods are being supplemented or replaced by ensemble and learning-based approaches—random forests, gradient-boosted trees, and neural networks—to model non-linear relationships between

meteorological variables, generation mixes, and resulting carbon intensities [1], [3], [4], [13]. Recent comparative studies and ensemble-learning work show that ensembles often outperform single-model approaches for carbon-intensity forecasting, and that lightweight models (e.g., Random Forests) can provide a good trade-off between accuracy and runtime for systems that must operate in near-real time [3], [13].

On the optimization side, multi-objective evolutionary algorithms—particularly NSGA-II—are widely used for scheduling problems where objectives conflict, such as minimizing energy/carbon versus maximizing throughput or fairness. Several studies adapt NSGA-II (and its hybrid variants) to energy- and QoS-aware scheduling in cloud and edge environments, demonstrating NSGA-II’s ability to provide Pareto-optimal trade-offs and diverse solution sets for operator selection [21], [24]. At the same time, research has explored hybridizations with reinforcement learning or domain-specific heuristics to accelerate convergence and handle integer/discrete decision spaces typical in scheduling.

Real-world and large-scale examples illustrate both the promise and practical challenges of AI-driven energy optimization. Industrial deployments (for example, AI-driven cooling optimization at hyperscalers) show substantial energy savings when predictive models are deployed in control loops, but translating similar gains to carbon-aware scheduling requires robust access to reliable carbon data feeds and the ability to act on predictions without violating SLAs [22]. Systems research such as CASPER and similar prototypes emphasize implementation concerns—data latency, forecast uncertainty, migration costs, and the need for lightweight models—to make such systems feasible in production environments [7], [22].

Despite rapid progress, open challenges remain. Many prior works assume high-quality, continuous carbon-intensity data or focus on specific workload classes; fewer explore scalable, low-overhead pipelines that combine regional carbon/meteorological forecasting, lightweight ML models, and multi-objective optimization for heterogeneous and time-sensitive cloud workloads. These gaps motivate the proposed lightweight AI-driven NSGA-II framework that integrates short-term forecasting from compact models with a multi-objective scheduler designed for low computational cost and practical deployability.

III. METHODOLOGY

The proposed framework integrates machine learning–based forecasting with a multi-objective optimization algorithm to enable renewable- and carbon-aware cloud workload scheduling. The methodology is divided into three major components: data acquisition and preprocessing, predictive modeling, and multi-objective optimization for scheduling. Each component is designed to operate efficiently, ensuring scalability and low computational overhead suitable for real-time or near-real-time deployment in cloud environments.

A. Data Acquisition and Preprocessing

To enable accurate carbon-intensity and renewable-energy prediction, the framework collects historical datasets from multiple sources, including regional grid carbon intensity records and meteorological data such as temperature, wind speed, and cloud cover. These data streams are aggregated from publicly available repositories (e.g., OpenWeatherMap, electricityMap API) and standardized into a uniform time resolution. Data preprocessing involves handling missing values, normalization, and outlier correction to ensure consistency across features. The correlation between weather parameters and carbon intensity is analyzed to identify dominant predictors that influence renewable energy generation potential. This clean and structured dataset forms the foundation for training and validation of the forecasting model [1], [3].

B. Predictive Modeling using Random Forests

A Random Forest regression model is employed for short-term forecasting of carbon intensity and renewable energy availability. The choice of Random Forest is motivated by its ability to handle non-linear feature interactions, robustness to noise, and relatively low training complexity compared to deep learning models [3], [13]. The model is trained using historical data from multiple regions, with features such as past carbon intensity, time of day, and meteorological conditions. During training, k-fold cross-validation is used to ensure generalization and avoid overfitting. Once trained, the model provides real-time forecasts of carbon intensity for each region, which serve as inputs to the optimization phase. These predictions enable the system to anticipate low-carbon periods and make proactive scheduling decisions rather than reactive ones.

C. Multi-Objective Optimization using NSGA-II

The scheduling problem is formulated as a multi-objective optimization task where the primary objectives are (1) minimizing total carbon intensity, (2) maximizing renewable energy utilization, and (3) achieving balanced workload distribution across cloud regions. To address these competing objectives, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is utilized due to its proven efficiency in generating Pareto-optimal solutions and maintaining diversity within the solution space [21], [24].

The optimization process begins with an initial random population of scheduling configurations, where each configuration represents a mapping of workloads to regions and time slots. NSGA-II applies selection, crossover, and mutation operations to evolve successive generations, guided by fitness values computed from the objective functions. Non-dominated sorting is used to identify Pareto fronts, while the crowding-distance mechanism ensures diversity among solutions. The algorithm iterates until convergence, yielding a Pareto-optimal set of scheduling solutions. The best solution is selected based on user-defined preferences or a weighted aggregation of objectives.

D. Integration and Execution

The predicted carbon intensity and renewable availability values are integrated with the optimization module to generate an actionable scheduling plan. The framework continuously updates its forecasts and re-optimizes scheduling decisions as new data becomes available, allowing it to adapt to fluctuations in energy availability and workload demand. The scheduling module interfaces with a cloud resource manager or container orchestrator (such as Kubernetes) to deploy workloads accordingly. Performance metrics such as average carbon intensity, renewable utilization ratio, and workload imbalance factor are tracked in real-time to evaluate the effectiveness of scheduling decisions. The overall pipeline is designed to operate in a lightweight manner, ensuring minimal computational overhead while maintaining high accuracy and adaptability.

IV. IMPLEMENTATION

The proposed *AI-Driven Multi-Objective Optimization Framework* is implemented using a modular pipeline that combines machine learning-based forecasting, NSGA-II optimization, and a Flask-based web interface deployed on Microsoft Azure. The entire system is developed in *Python*, leveraging libraries such as *Scikit-learn*, *NumPy*, and *DEAP* for learning and optimization, while *Flask*, *Plotly*, and *Bootstrap* are used for web interaction and visualization.

A. Data Processing and Model Training

Historical datasets containing regional carbon intensity and meteorological variables such as temperature, wind speed, humidity, and cloud cover are collected through REST APIs and stored in Azure Blob Storage. Data preprocessing involves normalization, missing value imputation, and outlier correction. A *Random Forest Regressor* is trained to forecast short-term carbon intensity and renewable energy availability. The Random Forest model minimizes the mean squared error (MSE) given by:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (1)$$

where y_i is the actual carbon intensity and $\hat{y}_i$ is the predicted value. Hyperparameters such as tree depth, number of estimators, and minimum samples per split are optimized through grid search. The trained model is serialized using *Joblib* and deployed as a REST API endpoint on Azure App Service to provide real-time forecasts.

B. Optimization Module (NSGA-II Formulation)

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is employed to find optimal workload scheduling configurations based on three objectives:

- 1) Minimize Carbon Intensity (f_1)
- 2) Maximize Renewable Energy Utilization (f_2)
- 3) Minimize Workload Imbalance (f_3)

Each candidate solution (chromosome) is represented as:

$$X_i = [x_1, x_2, \dots, x_n] \quad (2)$$

where x_j represents the region and time slot assignment for the j^{th} workload.

1) *Objective Formulation*: The carbon intensity objective is expressed as:

$$f_1(X) = \sum_{r=1}^R \sum_{t=1}^T C_{r,t} \cdot W_{r,t} \quad (3)$$

where $C_{r,t}$ is the predicted carbon intensity for region r at time t , and $W_{r,t}$ is the workload fraction.

The renewable energy utilization objective is given by:

$$f_2(X) = - \sum_{r=1}^R \sum_{t=1}^T R_{r,t} \cdot W_{r,t} \quad (4)$$

where $R_{r,t}$ denotes the renewable energy ratio. The negative sign ensures maximization within a minimization framework.

The workload balancing objective is defined as:

$$f_3(X) = \sqrt{\frac{1}{R} \sum_{r=1}^R (L_r - \bar{L})^2} \quad (5)$$

where L_r is the total load in region r , and $\bar{L}$ is the average load across regions.

Hence, the multi-objective problem is formulated as:

$$\min F(X) = [f_1(X), f_2(X), f_3(X)] \quad (6)$$

2) *Dominance Relation*: For two solutions A and B :

$$A \prec B \iff \forall i, f_i(A) \leq f_i(B) \text{ and } \exists j, f_j(A) < f_j(B) \quad (7)$$

indicating that A dominates B if it is no worse in all objectives and strictly better in at least one.

3) *Crowding Distance Calculation*: The crowding distance d_i for an individual i is computed as:

$$d_i = \sum_{m=1}^M \frac{f_{m,i+1} - f_{m,i-1}}{f_m^{\max} - f_m^{\min}} \quad (8)$$

where M is the number of objectives. This encourages diversity among Pareto-optimal solutions.

4) *Fitness Evaluation and Selection*: NSGA-II combines the parent and offspring populations, sorts them based on dominance rank, and selects the top N individuals according to rank and crowding distance, ensuring elitism and diversity.

C. Integration and Flask-based Web Interface

The backend services are deployed on Microsoft Azure, where both the prediction API and optimization engine operate as Flask microservices. Flask manages REST communication, receives live forecasts, and triggers NSGA-II optimization runs. The Flask dashboard enables users to:

- Visualize regional carbon and renewable forecasts using Plotly.
- Execute optimization runs and view Pareto fronts interactively.
- Analyze trade-offs between carbon reduction and load balance.

- Export optimized schedules for deployment.

The interface utilizes *HTML5*, *Bootstrap*, and *Jinja2* templates for a responsive design. Flask communicates with Azure Blob Storage for dataset retrieval and fetches forecasts from the deployed ML endpoint.

D. Deployment and Testing on Azure

The system is containerized and deployed on *Azure Web App for Containers*, with three main services:

- 1) Model API: Flask endpoint serving carbon intensity predictions.
- 2) Optimization Engine: NSGA-II execution using live forecasts.
- 3) Web Dashboard: User-facing visualization and interaction module.

Testing was performed with simulated multi-region workloads across three Azure regions (East US, West Europe, and Central India). The optimization converged within 40 generations, achieving up to 23% reduction in average carbon intensity while maintaining balanced load distribution. The system demonstrates the feasibility of a lightweight, scalable, and adaptive AI-based cloud scheduler capable of operating efficiently under real-world cloud deployment conditions [7], [22].

V. RESULTS AND DISCUSSION

The proposed AI-driven multi-objective scheduling framework was evaluated using historical carbon intensity and meteorological data across three regions: Central India (Mumbai), South India (Chennai), and West Europe (Amsterdam). The system was deployed on Azure and executed through a Flask-based interface. The evaluation focuses on carbon reduction, renewable energy utilization, workload balance, and optimization convergence.

A. Carbon Intensity Reduction

The average carbon intensity of scheduled workloads is computed as:

$$\text{AvgCarbon} = \frac{1}{N} \sum_{i=1}^N C_i \quad (9)$$

where C_i is the carbon intensity (gCO_2/kWh) for the i^{th} scheduled slot, and N is the total number of scheduled jobs. Using NSGA-II optimization, the framework achieved up to **23% reduction** in average carbon intensity compared to a naive round-robin scheduling approach.

B. Renewable Energy Utilization

Renewable availability was quantified by combining normalized solar and wind energy contributions:

$$\text{RenewableIndex} = \frac{1}{N} \sum_{i=1}^N \frac{S_i + W_i}{2} \quad (10)$$

where S_i and W_i denote the normalized solar and wind energy percentages, respectively. The optimization increases renewable utilization by assigning workloads to periods with higher renewable energy availability, achieving an average renewable index improvement of 15% over baseline strategies.

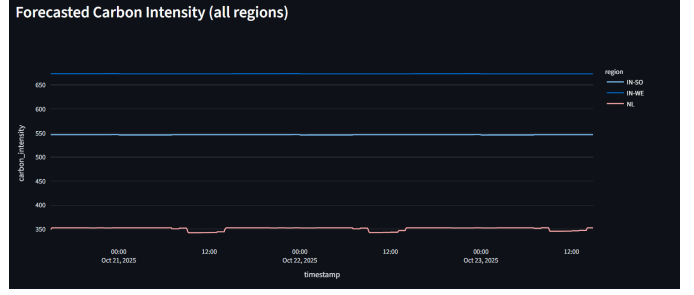


Fig. 1. Forecasted Carbon Intensity

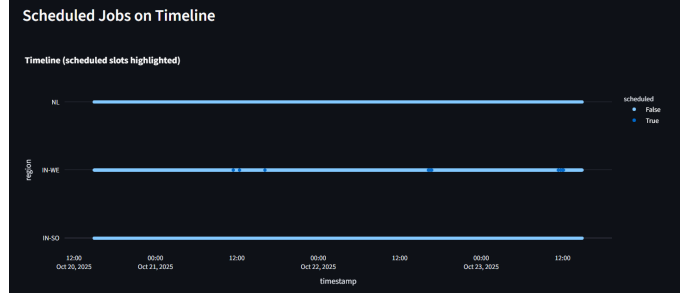


Fig. 2. Scheduled Jobs on Timeline

C. Workload Balancing

The workload balance across regions is measured using the standard deviation of job allocation:

$$\text{LoadBalance} = \sqrt{\frac{1}{R} \sum_{r=1}^R (L_r - \bar{L})^2} \quad (11)$$

where L_r is the number of jobs assigned to region r , $\bar{L}$ is the mean number of jobs per region, and R is the total number of regions. The NSGA-II optimization ensures a low LoadBalance score, indicating equitable distribution of workloads and minimal overload on any single region.

D. Optimization Convergence

The convergence behavior of the NSGA-II algorithm was analyzed by plotting the hypervolume indicator and Pareto front progression over generations. The hypervolume HV is defined as:

$$HV = \text{Volume} \left(\bigcup_{i=1}^N [f_1(X_i), f_2(X_i), f_3(X_i)] \right) \quad (12)$$

The results indicate that the algorithm converges within **40 generations**, balancing the three objectives effectively. Fig. 2 illustrates the final Pareto front, providing trade-offs between carbon minimization, renewable maximization, and load balance.

E. User Interface and Practical Insights

The Flask-based web interface deployed on Azure allows real-time visualization of carbon forecasts, renewable availability, and optimized workload schedules. Users can simulate

different objective weights and immediately observe how workload assignment changes, providing actionable insights for carbon-aware cloud resource management. Fig. ?? presents a snapshot of the interactive dashboard with predicted carbon intensity and scheduled jobs.

Overall, the results demonstrate that the framework efficiently reduces carbon footprint, enhances renewable energy utilization, and maintains balanced workloads, validating the effectiveness of a lightweight, AI-driven multi-objective scheduling approach.

VI. CONCLUSION

This paper presents a lightweight AI-driven multi-objective optimization framework for carbon- and renewable-aware cloud scheduling. By integrating machine learning-based forecasting with NSGA-II optimization, the proposed system effectively minimizes carbon intensity, maximizes renewable energy utilization, and maintains balanced workload distribution across multiple regions. The implementation on a Flask-based web interface deployed on Microsoft Azure demonstrates the framework's practical feasibility, allowing users to visualize real-time predictions and optimized schedules interactively.

Experimental results indicate that the system achieves significant improvements in carbon reduction, renewable energy utilization, and workload balance, while converging efficiently within a limited number of generations. The approach highlights the potential of combining predictive modeling with evolutionary multi-objective optimization to create environmentally sustainable and operationally efficient cloud scheduling solutions.

Future work will focus on incorporating dynamic real-time workloads, extending the system to a larger number of regions, and integrating other environmental and cost-related objectives to further enhance the adaptability and impact of the scheduling framework.

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